Lok Jagruti Kendra University
Ahmedabad – 382210

Course Code	**140110208**			
Category	**Core Subject**			
Course Title	**Python Programming**			
Scheme and Credits	**Theory**	**Tutorial**	**Lab**	**Credits**
	0	**2**	**4**	**4**
Pre-requisites (if any)	Knowledge of the C programming language			

1.Course Objectives:

1	To understand Fundamental of Python Programming
2	To Understand the Flow Control and Functions of Python Programming
3	To understand File operations, Database, and exception handling in Python
4	To implement, Plot and Visualize the Graphs using Pandas, and matplotlib

2. Course Contents:

Unit	Course Content	Hours	Weightage
Unit I	• History & need of Python • Application of Python • Advantages of Python • Disadvantages of Python • Installing Python • Program structure • Interactive Shell • Executable or script files. • User Interface or IDE PYTHON FUNDAMENTALS • Working with Interactive mode • Working with Script mode • Python Character Set • Python Tokens, Keywords, Identifiers, Operators • Variables and Assignments • Input and Output in Python DATA HANDLING • Data Types Numbers , Strings , Lists , Tuples , Dictionary, Mutable and Immutability	**20 Hours** **6** **10** **4**	**25%**
Unit II	PROGRAM CONTROL FLOW • Conditional Statements The if Statement ,The if-else Statement , The if-elif Statement , Nested if Statements , Python Indentation • Looping and Iteration , The For Loop , The While Loop , Loop else Statement , Nested Loops , Break and Continue • The Range Function , Introduction to range() , Types of range() function , Use of range() function INTRODUCTION TO FUNCTIONS • Built-In Functions Introduction to Functions , Using a Functions , Python Function Types , Structure of Python Functions ,	**20 Hours** **10** **10**	**25%**

	• User Defined Functions , Structure of a Python Program w.r.t. UDF , Types of Functions , Invoking UDF , Flow of Execution , Arguments and Parameters , Default Arguments, Named Arguments , Scope of Variables , Lambda function • Recursion Function , Use of recursion function		
Unit III	FILE OPERATIONS • Text and Bytes files , Opening a file , Reading and Writing Files , Other File tools • MS Excel files , (.xls,.csv) DATABASE • Introduction to MySQL • PYMYSQL Connections • Executing queries • Transactions • Handling error EXCEPTION HANDLING • Default Exception and Errors • Catching Exceptions • Raise an exception • Try.... except statement • Raise, Assert, Finally blocks • User defined exception	**15 Hours** **5** **5** **5**	**25%**
Unit IV	Pandas**:** Manipulating data from CSV, Excel, HDF5, and SQL databases, • Using multilevel series • Series and Data Frames • Grouping, aggregating • Merge DataFrames • Generate summary tables • Group data into logical pieces • Manipulate dates • Creating metrics for analysis • Data wrangling • Merging and joining Visualization with matplotlib: • Scatter plot • Bar charts, histogram • Stack charts • Legend title Style • Figures and subplots • Plotting function in pandas • Labelling and arranging figures • Save plots	**17 Hours** **7** **10**	**25%**

3. Text Books:

- Core python programming by Dr. R. Nageshwar Rao
- Pandas for Everyone by Daniel Y. Chen

4. Reference book:

- McKinney Wes, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython", O'Reilly Media, 2012.
- Hauck Trent, "Instant Data Intensive Apps with Pandas How-To", Packt Publishing Ltd, 2013.

4. Webilography:

1. https://www.python.org
2. https://www.w3schools.com/python

5. Tools: Latest Version of IDLE, PyCharm

6. Accomplishment of the student after completing the course:

After completion of the course students should be capable of developing console-based application through Python programming language with Plot and Visualize the Graphs for Data analysis.

Unit	Book	Chapter
I	Core python programming by Dr. R. Nageshwar Rao	Chapter 1, Chapter 3, Chapter 4, Chapter 5
II	Core python programming by Dr. R. Nageshwar Rao	Chapter 6, Chapter 9
III	Core python programming by Dr. R. Nageshwar Rao	Chapter 16, Chapter 17, Chapter 24
IV	Pandas for Everyone by Daniel Y. Chen	Chapter 1, Chapter 2, Chapter 3, Chapter 4